

Name: _____

Instructions:

- DO NOT TURN TO THE SECOND PAGE UNTIL YOUR INSTRUCTOR HAS ALLOWED YOU TO DO SO.
- You are allowed a one-side page cheat sheet, but no other notes or resources. You are NOT allowed to use any other external resource like cellphones, lecture notes, books or talk to any of your peers.
- Show your work. Full points are only given for correct answers *with* adequate justification. A correct final answer with missing or substantially incorrect justification will not merit full points on a problem.
- If you need an extra page to complete your computations for a problem you will find another page at the end of this midterm. Indicate clearly if you used this extra space while solving a problem, for example leaving a message on the page of Problem 3: *the rest of my solution to Problem 3 is in the extra-page.*
- Good luck! Read the entire midterm first and start with those points that make you feel more confident.

Problem 1. (1 pts)

Let $(G, \star, e)$ be a group and $H \leq G$ be a subgroup. Given $g \in G$, let

$$\begin{aligned}\mu : G \times G &\rightarrow G \\ (g, h) &\mapsto g^{-1}hg\end{aligned}$$

1. State the definition of an action of a group G on a set S .
2. Show that μ is an action of G on G .
3. Show that if G is abelian then the action μ on G is trivial, i.e. for fixed $g \in G$, $\mu(g, h) = h$ for all $h \in G$.
4. Prove that for any $g \in G$, the map $\mu_g : G \mapsto G$ that sends an element h to $g^{-1}hg$ is an automorphism.
5. Let $H = \{\mu_g \mid g \in G\}$ show that H is a subgroup of $(\text{Aut}(G), \circ, Id_G)$

Problem 2. Let $(G, \cdot, e)$ be a group and let H be a subgroup.

1. Consider the relation on G defined by $g_1 \sim g_2$ if and only if $g_1^{-1}g_2 \in H$. Prove that it is an equivalence relation.
2. We say that H induces 2 classes in G if there are $g_1, g_2 \in G$ such that for every $h \in G$ we have that $g_1 \sim h$ or $g_2 \sim h$.
3. Show that having a subgroup with 2 classes is a structural property.
4. Can you provide an example?

Problem 3. Let $(G, \cdot, e)$ and $(H, \star, e)$ be groups.

1. We will construct a group operation on the cartesian product. Define a binary operation as:

$$\begin{aligned}\phi : (G \times H) \times (G \times H) &\mapsto G \times H \\ (g_1, h_1), (g_2, h_2) &\rightarrow (g_1 \cdot g_2, h_1 \star h_2)\end{aligned}$$

Show that $(G \times H, \phi)$ is a group.

2. Let H be a subgroup of G .

- Show that for $g \in G$ then gHg^{-1} is also a subgroup of G .
- H is said to be normal if $gHg^{-1} = H$ for all $g \in G$.
- Let $G/H = \{[g]_{\sim} \mid g \in G\}$ where $\sim$ is the relation on G defined as $g_1 \sim g_2$ if and only if $g_1^{-1} \cdot g_2 \in H$. Consider the operation:

$$\begin{aligned}\star : G/H \times G/H &\mapsto G/H \\ ([g_1], [g_2]) &\rightarrow [g_1 g_2]\end{aligned}$$

Show that $(G/H, \star)$ is a group when H is a normal subgroup.

Problem 4.

1. Classify all subgroups of $\mathbb{Z}_{98} = \mathbb{Z}/98\mathbb{Z}$.
2. Assume that p and q are different prime numbers. Consider $(\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}, \phi)$ with the group operation defined in problem 3.1. Show that $(\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}, \phi)$ is isomorphic to $\mathbb{Z}/pq\mathbb{Z}$.

Problem 5. Let $(G, \cdot, e_G)$ and $(H, \star, e_H)$ be groups. Let $\phi : G \rightarrow H$ be an homomorphism. Let $\text{Ker}(\phi) = \{g \in G \mid \phi(g) = e_H\}$.

1. Show that $\text{Ker}(\phi)$ is a subgroup of G ,
2. Show that $\text{Ker}(\phi)$ is a normal subgroup,
3. Is the converse true? Is every normal subgroup the Kernel of some homomorphism?
(This question might be too hard for the stage we are at, but so that you get inspired.)